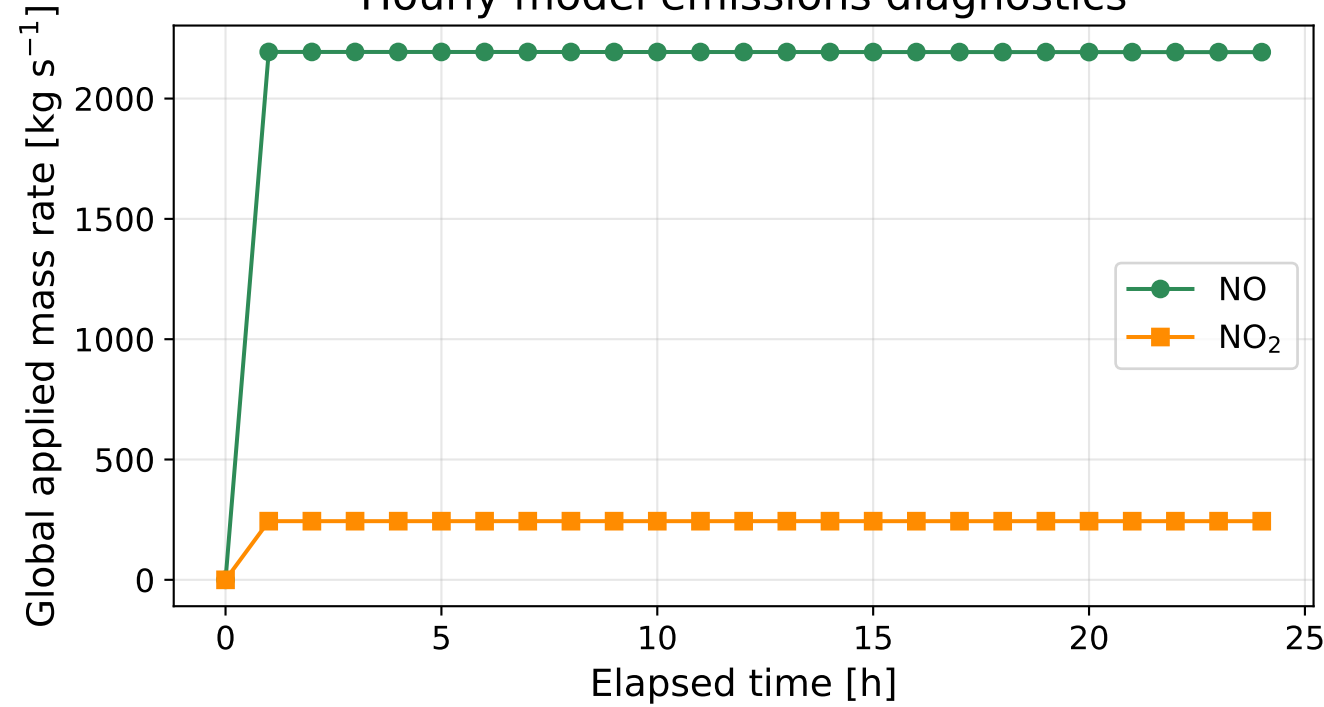


A1 global MIEM source, Chapman-NOx chemistry, and matched-control response

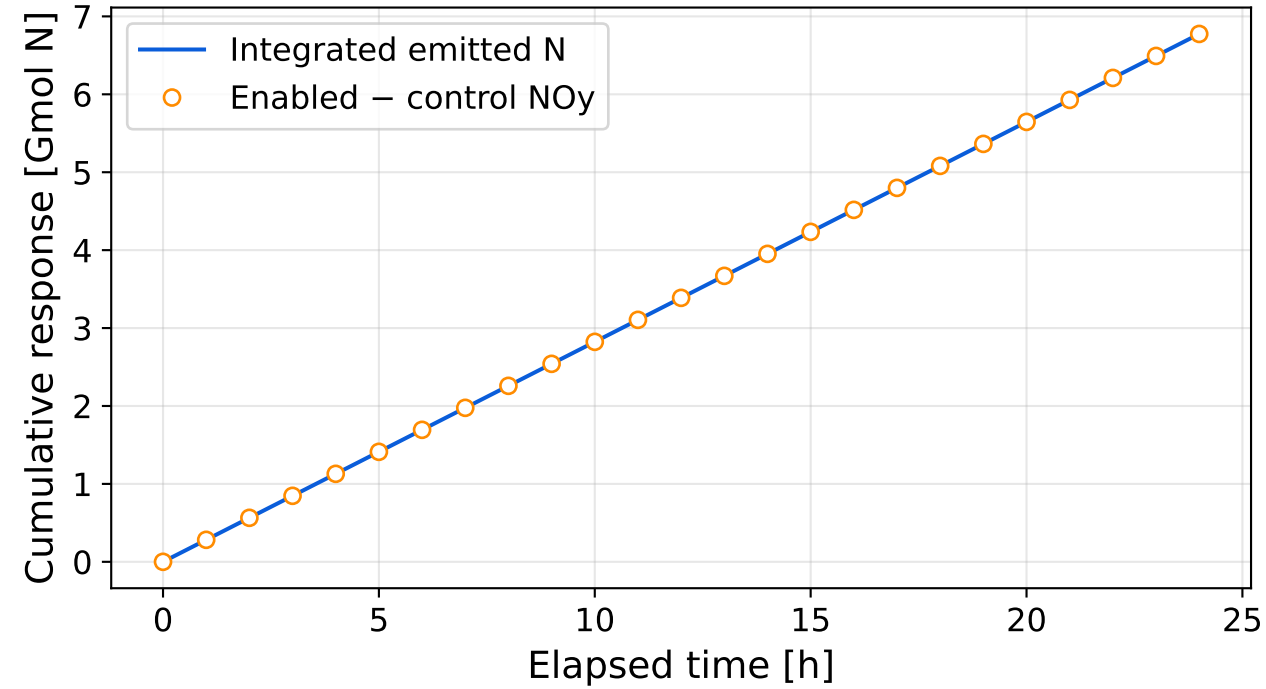
24 h continuous trajectory | 8 MPI ranks | lightning and other N sources disabled

Hourly model emissions diagnostics

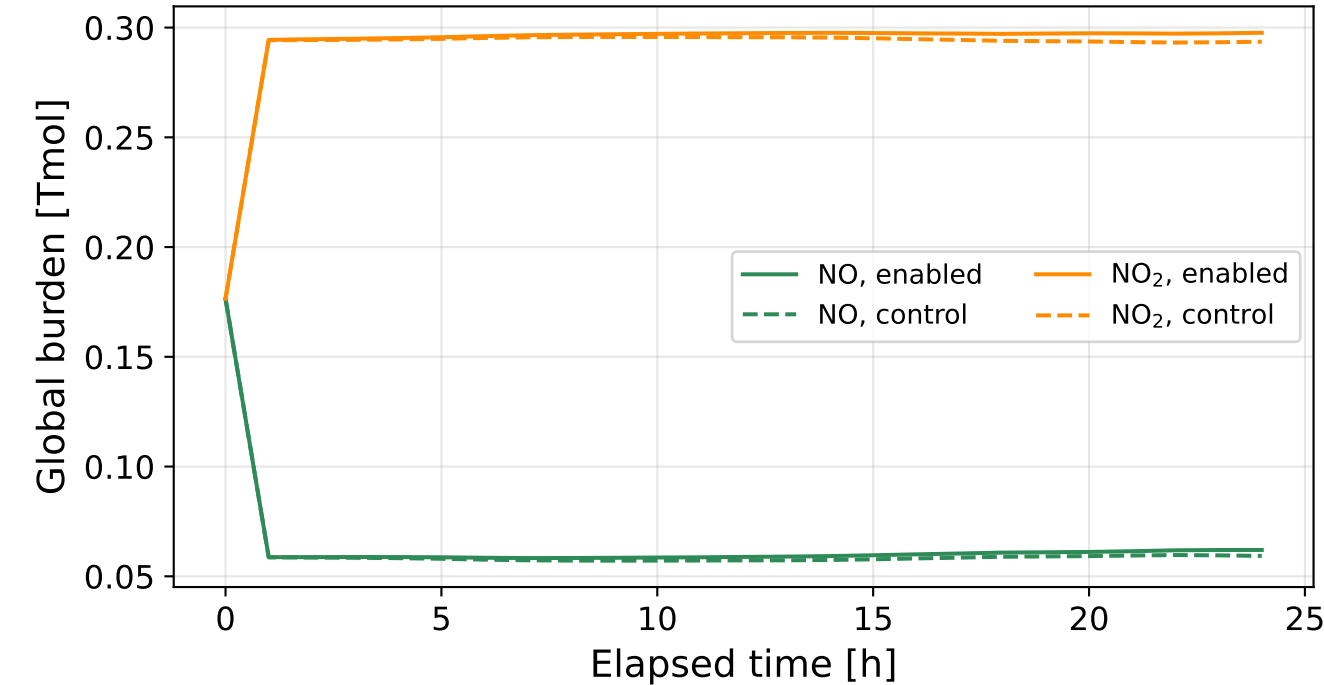


Reactive-N source/response closure

final relative residual = $1.44\text{e-}11$



Photochemical NO/NO2 partitioning



Retained sector diagnostics at 24 h

